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# PROJECT REPORT
# ON
# Stock Market Analysis Using Microsoft Excel

Program Name: BCA

Subject Name/Code: Data Science Lab Using Excel

(25CAP-122)


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# Project Report: Stock Market Analysis Using Microsoft Excel

**1. Title:-** Stock Market Analysis Using Microsoft Excel

## 2. Aim of the Project:-

The aim of this project is to design and develop a comprehensive and automated Excel-based system that enables efficient tracking, evaluation, and visualization of stock market performance data across multiple companies and sectors listed on the National Stock Exchange (NSE) of India. The system is built to handle 100 real-world inspired trading entries covering 20 major companies spread across 12 different sectors.

The project focuses on reducing manual data entry and calculation effort by leveraging Excel's powerful built-in formula engine. By integrating formulas, conditional formatting, pivot tables, and 9 interactive analytical charts, the system provides a complete financial data analysis environment. The goal is to offer both beginners and experienced data analysts an intuitive platform to assess market trends, sector performance, and individual stock behavior — all within Microsoft Excel.

## 3. Objective:-

The primary objective of this project is to record, manage, and analyze stock market performance data for 100 trading entries across 20 NSE-listed Indian companies, providing analysts and investors with a quick and reliable way to evaluate market trends and company-level performance.

Additional objectives include:

- To reduce manual work and improve efficiency in financial data evaluation.
- To use Excel formulas such as AVERAGE(), SUM(), IF(), RANK(), and COUNTIF() to automate repetitive financial calculations.
- To enhance stock market result analysis through graphical representation using 9 different chart types.
- To help investors and financial analysts identify high-performing sectors, top-volume stocks, and price range leaders.
- To provide a professional, well-structured financial analysis tool suitable for academic and institutional use.
- To demonstrate practical data science skills using Excel as the primary analytical tool.

## 4. Tools and Technologies Used:-

The following tools, formulas, and Excel techniques were used throughout the development of this project:

**Software Tools:**

- Microsoft Excel 2016 or higher (primary development platform)
- Microsoft Windows 10 / 11 operating environment
- GitHub for version control and project file hosting

**Excel Formulas Used:**

- AVERAGE() — To compute average closing price, average volume, and average P/E ratio per company and sector.
- SUM() — To calculate total trading volume across all entries.
- IF() — To apply logical conditions for flagging gains versus losses.
- COUNT() / COUNTA() — To count total entries and non-empty cells.
- COUNTIF() — To count positive and negative trading days.
- MAX() / MIN() — To identify the highest and lowest closing prices.
- RANK() — To rank companies by performance metrics.

**Excel Features Used:**

- Conditional Formatting — For color-coding gains (green) and losses (red) in the Change columns.
- Bar Charts, Line Charts, Pie Charts, Area Charts, Clustered Bar Charts, and Stacked Bar Charts — For visual performance analysis.
- Pivot Tables — For sector-level summary and class-level analysis.
- Freeze Panes — To keep header rows visible while scrolling through 100 entries.
- Data Validation — To ensure consistent data entry across all rows.

## 5. System Requirements:-

To ensure smooth performance and compatibility, the following hardware and software configurations were used and are recommended for running this project:

**Hardware Requirements:**

- A standard computer or laptop with at least 4 GB RAM
- Processor: Intel Core i3 (3rd Gen or higher) or equivalent AMD processor
- Storage: Minimum 500 MB free disk space for data files, charts, and backups
- Display: 1366 x 768 or higher resolution for comfortable chart viewing

**Software Requirements:**

- Operating System: Windows 10 / Windows 11 or macOS Monterey or later
- Primary Software: Microsoft Excel 2016, 2019, 2021, or Microsoft 365
- Optional: LibreOffice Calc (as a free alternative for viewing the workbook)
- Web Browser: Any modern browser to access GitHub repository

## 6. Algorithm / Logic:-

The Stock Market Analysis System follows a structured, step-by-step computational process to transform raw trading data into a fully functional analytical dashboard. The logic is divided into three phases: Data Entry, Computation, and Visualization.

**Phase 1 — Data Entry & Setup:**

**1.** Create a new Excel workbook with three sheets: 'Stock Market Data', 'Summary', and 'Charts & Analysis'.

**2.** In the 'Stock Market Data' sheet, define 14 column headers: S.No, Company Name, Ticker, Sector, Date, Open (₹), High (₹), Low (₹), Close (₹), Volume, Market Cap (Cr), P/E Ratio, Change (₹), and Change (%).

**3.** Enter 100 rows of stock trading data for 20 NSE-listed companies across 12 sectors. Each company appears 5 times, representing 5 different trading dates from January to June 2025.

**4.** Apply freeze panes at row 3 to keep headers visible while scrolling.

**Phase 2 — Formula Computation:**

**5.** In the Change (₹) column, apply the formula =I3-F3 for every row to compute daily price change.

**6.** In the Change (%) column, apply the formula =M3/F3 to calculate the percentage movement relative to the opening price.

**7.** Apply Conditional Formatting rules: green background for positive change values and red background for negative change values.

**8.** In the 'Summary' sheet, use AVERAGE(), MAX(), MIN(), SUM(), and COUNTIF() formulas to pull statistics from the data sheet using cross-sheet references.

**Phase 3 — Visualization:**

**9.** In the 'Charts & Analysis' sheet, build 5 data tables aggregating: sector-wise averages, company-wise volumes, P/E by sector, price ranges, and positive/negative day counts.

**10.** Generate 9 charts linked to those data tables: Avg Close by Sector (Column Bar), Total Volume by Company (Column Bar), Entries by Sector (Pie), Close Price Trend (Line), P/E by Sector (Horizontal Bar), Price Range per Company (Clustered Bar), Positive vs Negative Days (Stacked Bar), Volume Trend (Area), and Avg Close per Company (Column Bar).

**11.** Format all charts with appropriate titles, axis labels, and color schemes for professional presentation.

## 7. Code Overview (Excel Formulas):-

The following Excel formulas form the computational backbone of the Stock Market Analysis System. Each formula is designed to be dynamic and automatically recalculate whenever source data is updated.

### Core Data Formulas (Applied to Each Row):

```
=I3-F3
```

→ *Calculates daily price change (Close Price minus Open Price). Applied to all 100 rows in the Change (₹) column (Column M).*

```
=M3/F3
```

→ *Calculates daily percentage change relative to opening price. Applied to all 100 rows in Change (%) column (Column N). Formatted as percentage.*

### Summary Sheet Formulas (Cross-Sheet References):

```
=COUNTA('Stock Market Data'!A3:A102)
```

→ *Counts total number of data entries in the tracker (should return 100).*

```
=AVERAGE('Stock Market Data'!I3:I102)
```

→ *Calculates the average closing price across all 100 trading entries.*

```
=MAX('Stock Market Data'!I3:I102)
```

→ *Returns the highest closing price recorded across all entries.*

```
=MIN('Stock Market Data'!I3:I102)
```

→ *Returns the lowest closing price recorded across all entries.*

```
=SUM('Stock Market Data'!J3:J102)
```

→ *Calculates total cumulative trading volume across all 100 entries.*

```
=AVERAGE('Stock Market Data'!L3:L102)
```

→ *Computes average Price-to-Earnings (P/E) ratio across all entries.*

```
=COUNTIF('Stock Market Data'!N3:N102,">0")
```

→ *Counts the number of entries where daily percentage change was positive (bullish days).*

```
=COUNTIF('Stock Market Data'!N3:N102,"<0")
```

→ *Counts the number of entries where daily percentage change was negative (bearish days).*

```
=AVERAGE('Stock Market Data'!N3:N102)
```

→ *Computes the average daily percentage change — a key indicator of overall market trend.*

### Conditional Formatting Logic:

Two conditional formatting rules are applied to columns M and N (Change ₹ and Change %):

- Rule 1: If cell value > 0, apply GREEN fill (RGB: 214, 244, 214) and dark green bold text — indicating a profitable trading day.
- Rule 2: If cell value < 0, apply RED fill (RGB: 253, 222, 222) and dark red bold text — indicating a loss-making trading day.
- This visual differentiation allows analysts to instantly identify bullish and bearish entries without manual inspection.

## 8. Dataset Description:-

The dataset used in this project contains 100 simulated but realistic stock trading entries inspired by actual NSE-listed Indian companies. The data spans trading dates from January 2025 to June 2025 and covers 20 companies across 12 market sectors.

**Companies and Sectors Covered:**

| S.No | Company Name | Ticker | Sector |
|---|---|---|---|
| 1 | Reliance Industries | RELIANCE | Energy |
| 2 | Tata Consultancy Services | TCS | IT |
| 3 | HDFC Bank | HDFCBANK | Banking |
| 4 | Infosys | INFY | IT |
| 5 | ICICI Bank | ICICIBANK | Banking |
| 6 | Hindustan Unilever | HINDUNILVR | FMCG |
| 7 | ITC Ltd | ITC | FMCG |
| 8 | Larsen & Toubro | LT | Infrastructure |
| 9 | State Bank of India | SBIN | Banking |
| 10 | Bajaj Finance | BAJFINANCE | Finance |
| 11 | Wipro | WIPRO | IT |
| 12 | Asian Paints | ASIANPAINT | Materials |
| 13 | Maruti Suzuki | MARUTI | Auto |
| 14 | Sun Pharma | SUNPHARMA | Pharma |
| 15 | Titan Company | TITAN | Consumer |
| 16 | Kotak Mahindra Bank | KOTAKBANK | Banking |
| 17 | Adani Ports | ADANIPORTS | Infrastructure |
| 18 | Power Grid Corp | POWERGRID | Utilities |
| 19 | NTPC Ltd | NTPC | Utilities |
| 20 | Bharti Airtel | BHARTIARTL | Telecom |

**Data Columns Explained:**

- S.No — Serial number (1 to 100)
- Company Name — Full name of the NSE-listed company
- Ticker — NSE trading symbol (e.g., RELIANCE, TCS)
- Sector — Market sector classification (Banking, IT, FMCG, etc.)
- Date — Trading date in DD-MMM-YYYY format
- Open (₹) — Stock price at market open
- High (₹) — Intraday highest trading price
- Low (₹) — Intraday lowest trading price
- Close (₹) — Final price at market close
- Volume — Number of shares traded that day
- Market Cap (Cr) — Company market capitalization in Indian Crores
- P/E Ratio — Price-to-Earnings ratio for valuation reference
- Change (₹) — Formula-calculated daily price change
- Change (%) — Formula-calculated daily percentage change

## 9. Analysis Charts Description:-

The 'Charts & Analysis' sheet of the workbook contains 9 professionally designed interactive charts. Each chart is linked to a dedicated data table within the same sheet and updates automatically when source data changes.

| Chart # | Title | Chart Type | Purpose / Insight |
|---|---|---|---|
| 1 | **Avg Close by Sector** | Column Bar | Compares average closing stock price across all 12 sectors to identify which sectors have the highest-priced stocks. |
| 2 | **Total Volume by Company** | Column Bar | Shows total trading volume per company across all entries, identifying the most actively traded stocks. |
| 3 | **Entries by Sector** | Pie Chart | Displays the percentage distribution of data entries across sectors, showing sector representation in the dataset. |
| 4 | **Close Price Trend** | Line Chart | Tracks the closing price movement across the first 30 sequential entries, revealing short-term price trends. |
| 5 | **Avg P/E by Sector** | Horizontal Bar | Compares average Price-to-Earnings ratios by sector to assess relative valuations across the market. |
| 6 | **Price Range per Company** | Clustered Bar | Side-by-side comparison of Avg High, Avg Low, and Avg Close for each company — shows volatility range. |
| 7 | **Positive vs Negative Days** | Stacked Bar | Stacked view of bullish vs bearish trading days per company, giving a quick sentiment overview. |
| 8 | **Volume Trend** | Area Chart | Area visualization of trading volume across the first 30 entries to show volume spikes and patterns. |
| 9 | **Avg Close per Company** | Column Bar | Ranks all 20 companies by their average closing price, enabling direct price comparison across the portfolio. |

## 10. Screenshots / Output:-

The final output of the project is a professional, multi-sheet Excel workbook. The key outputs delivered by the system are described below:

**Sheet 1 — Stock Market Data:**

- 100 rows of complete trading data for 20 NSE companies.
- Auto-calculated Change (₹) and Change (%) columns with green/red conditional formatting.
- Freeze panes, professional color-coded headers, and alternating row shading.

**Sheet 2 — Summary Statistics:**

- 10 key metrics calculated via cross-sheet formulas: Total Entries, Avg/Max/Min Close Price, Total Volume, Avg Volume, Avg P/E Ratio, Positive Days, Negative Days, and Avg Daily Change %.

**Sheet 3 — Charts & Analysis:**

- 5 source data tables aggregating sector-level and company-level statistics.

- 9 interactive charts linked to data tables for dynamic visual analysis.
- All charts auto-update when the source data in Sheet 1 is modified.

**Summary Output Table:**

| Metric | Value (Approx.) |
|---|---|
| **Total Data Entries** | 100 |
| **Total Companies Covered** | 20 |
| **Total Sectors Covered** | 12 |
| **Average Closing Price** | ₹ 2,850 (approx.) |
| **Highest Closing Price** | ₹ 11,200+ (Maruti Suzuki) |
| **Lowest Closing Price** | ₹ 310 (NTPC / Power Grid) |
| **Average Daily Volume** | ~5.2 Million shares |
| **Average P/E Ratio** | ~34.5 |
| **Positive Trading Days** | ~52 out of 100 |
| **Negative Trading Days** | ~48 out of 100 |
| **Total Charts Generated** | 9 |
| **Total Formula Cells** | ~300+ |

## 11. Key Findings & Observations:-

Upon analyzing the 100-entry stock market dataset, the following key findings were observed:

- IT and Banking sectors recorded the highest average P/E ratios, indicating premium market valuations — consistent with India's growth-driven technology and financial services sectors.
- Maruti Suzuki (Auto sector) had the highest average closing price among all 20 companies, reflecting its premium positioning in the Indian passenger vehicle market.
- NTPC and Power Grid Corp had the lowest average prices but recorded consistently high trading volumes, suggesting strong institutional interest in utility stocks.
- Bharti Airtel (Telecom) showed significant volatility in the Price Range chart, with notable gaps between High and Low prices across trading sessions.
- Bajaj Finance (Finance) demonstrated one of the highest P/E ratios, consistent with its premium growth valuation in the NBFC (Non-Banking Financial Company) category.
- Approximately 52% of trading entries were positive (bullish), suggesting a mildly positive market sentiment across the observed period.
- The Volume Trend chart revealed that volume spikes correspond closely with entries showing larger price changes, confirming the volume-price relationship in equity markets.
- HDFC Bank, ICICI Bank, and SBI together dominated Banking sector volume, making Banking the highest-volume sector in the dataset.

## 12. Project Link:-

The complete Excel workbook for this project, including all 3 sheets (Stock Market Data, Summary, and Charts & Analysis), is hosted on GitHub and available for download at the following link:

https://github.com/vidurdvr/Data-Interpretation-23CAP-357-Mini-Project

## 13. Conclusion:-

This project successfully demonstrates how Microsoft Excel can be used as a powerful, accessible, and professional tool for automating and simplifying stock market data management and analysis. By combining structured data entry, dynamic Excel formulas, conditional formatting, pivot tables, and nine interactive data visualization charts, the Stock Market Analysis System delivers a comprehensive financial analysis environment within a single Excel workbook.

The project covers 100 trading entries across 20 major NSE-listed Indian companies spanning 12 diverse sectors — providing a realistic and representative sample of the Indian equity market. All formulas are dynamic, meaning the entire dashboard updates automatically whenever source data is modified, making the system scalable and reusable for future datasets.

Through this project, it is evident that Excel is not just a spreadsheet tool but a complete data analysis platform capable of handling real-world financial data with efficiency, accuracy, and visual clarity. The Stock Market Analysis System saves significant time for analysts, reduces manual calculation errors, and improves transparency and interpretability in financial evaluation — making it a valuable tool for both academic and professional contexts.

## 14. Learning Outcomes:-

Upon completing this project, the following skills and competencies were acquired:

- Acquired in-depth knowledge of Excel formulas including AVERAGE(), SUM(), IF(), COUNTIF(), MAX(), MIN(), and RANK() and their real-world financial applications.
- Learned the practical use of Conditional Formatting to highlight stock market gains and losses for rapid visual analysis.
- Understood how to build cross-sheet formula references to create linked Summary dashboards that pull data from master data sheets.
- Gained hands-on experience creating 9 different chart types (Bar, Line, Pie, Area, Clustered, Stacked) for multi-dimensional data analysis.
- Developed the ability to design and structure a professional multi-sheet Excel workbook suitable for academic submission and business reporting.
- Enhanced data presentation, financial reasoning, and analytical thinking skills applicable to data science and business intelligence roles.
- Understood the relationship between trading volume and price change — a core concept in equity market analysis.
- Improved GitHub usage skills for version control and project hosting.

## 15. References:-

The following resources were consulted during the research, development, and documentation of this project:


1. Microsoft Excel Official Documentation — https://support.microsoft.com/excel
2. TutorialsPoint — Excel Data Analysis Tutorial — https://www.tutorialspoint.com/excel
3. GeeksforGeeks — Excel Formula Guide — https://www.geeksforgeeks.org/ms-excel
4. YouTube — Excel Dashboard Tutorials by Chandoo.org
5. W3Schools — Excel Function Reference — https://www.w3schools.com
6. NSE India — National Stock Exchange Official Data — https://www.nseindia.com
7. Investopedia — Stock Market Terminology and Concepts — https://www.investopedia.com
8. GitHub — Project Repository — https://github.com/chauhansanchit723-wq/excel_workbook


## 16. Glossary of Terms:-

The following financial and technical terms are used throughout this report:

| Term | Definition |
|---|---|
| **OHLCV** | Open, High, Low, Close, Volume — the five fundamental data points of a stock trading session. |
| **NSE** | National Stock Exchange of India — the leading stock exchange platform in India. |
| **P/E Ratio** | Price-to-Earnings Ratio — a valuation metric comparing stock price to per-share earnings. |
| **Market Cap** | Market Capitalization — total market value of a company's outstanding shares, expressed in Crores (₹). |
| **Ticker** | A unique short code (symbol) used to identify a publicly traded company on a stock exchange. |
| **Bullish** | A market condition where prices are rising or expected to rise. |
| **Bearish** | A market condition where prices are falling or expected to fall. |
| **Conditional Formatting** | An Excel feature that automatically applies formatting (color, font) based on cell values. |
| **Pivot Table** | An Excel data summarization tool that aggregates and reorganizes data for analysis. |
| **NBFC** | Non-Banking Financial Company — a financial institution that provides banking services without a banking license. |